

realisation was that we were often designing without knowing what we didn't know. Critical concepts such as RAM-to-CPU routing, high-speed signal integrity, and power distribution networks weren't part of our initial thinking—they had to be learned along the way through online resources, courses, and continuous experimentation.”

For manufacturing, the startup currently relies on a contract facility in Baroda, near Surat, as it does not yet have its own SMT pick-and-place line. However, there is a clear plan to establish an in-house manufacturing setup within the next four months, beginning with the Shrike series and then gradually expanding production capacity for Axon and Vaaman over the following year. The company continues to handle component sourcing, debugging, testing, firmware flashing, and final validation independently, while packaging is also managed internally. As a result, only about 30%



Akshar Vastarpara, Founder and CEO, Vicharak

of the overall process is outsourced, with the remaining 70% retained and controlled in-house.

When asked about the current challenges the company is facing and addressing, Akshar discloses, “Hardware startups in India face a funding gap compared to D2C and product companies, which makes early growth difficult even with strong ideas. There is also a

talent gap—not in numbers, but in training for real hardware development, where engineers need more hands-on experience to build, test, and learn from real systems. At the same time, India has a huge base of software developers, and the real opportunity lies in giving them simple, accessible platforms to move into hardware and edge computing.”

Talking about the revenue generated by the company, Akshar says, “We were at a very pre-revenue stage for the last two years, during which we primarily sold beta kits and onboarded users onto the Vicharak platform. During this phase, we sold around 500 Axon units, nearly 100 Vaaman boards, and over 5000 Shrike-lite units. With a growing order pipeline and planned production expansion, we are targeting sales of at least 10,000 Axon and Vaaman units combined, along with around 50,000 Shrike-series units in the coming cycle.”

—Nidhi Agarwal

2 PUNE-BASED SLIGHTGEN SOLUTIONS POWERING INDIA'S PCB AUTOMATION WITH INDIGENEOUS ROBOTICS

Can manufacturers access cost-effective, customisable robotic soldering systems for PCB assembly automation? This Pune-based startup aims to address that challenge.

SlightGen Solutions LLP is a Maharashtra-based startup founded in 2024 by Govardhan Gondchawar and Shrikrushna Kakade. The Indian technology company develops indigenous automation solutions for electronics manufacturing, with robotic PCB soldering systems serving as its flagship product.

The company initially focused on embedded systems, IoT devices, data loggers, and biomedical products before shifting towards automation to address a clear gap in the market.

Nikesh Gondchawar, Strategic & Technical Advisor – Robotics

Manufacturing at SlightGen Solutions, explains that existing soldering solutions are either expensive



SG300x

imported machines from countries such as Japan, China, and Korea or low-cost alternatives that lack reliability. SlightGen positions itself as a high-value, locally engineered automation solution tailored for MSMEs.

The startup designs and develops its systems in-house, including software, electronics, and control systems, while mechanical fabrication and manufacturing are handled by an experienced manufacturing partner. System integration, controls, programming, firmware development, calibration, and final validation are performed internally by SlightGen, which has around